

Qualification Pack



Mine Mechanic/Fitter

Underground Metal/ Opencast/ Underground Coal

Repair-Underground/ Crusher

QP Code: MIN/Q3203

Version: 3.0

NSQF Level: 3.0

SCMS (Skill council of Mining Sector) || FIMI House, B-311, Okhla Industrial Area, Phase-I
New Delhi-110020 || email:scms@skillcms.in

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MIN/Q3203: Mine Mechanic/Fitter

Brief Job Description

A Mine Mechanic or Fitter in a mine is responsible for repair and maintenance of plant and machinery, medium / light vehicles, drilling machines, pumps, compressors, pneumatic machines, overhaul of crushers and other mechanical equipments and assemblies. A mechanic/fitter most often works with / assists senior technicians.

Personal Attributes

This job requires analytical thinking, ability to apply theory to practical situations, quality consciousness, safety orientation, physique to sustain strenuous conditions, ability to use fingers, hands and feet with ease to complete the assigned task (Dexterity).

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [MIN/N3211: Perform installation and preventive maintenance of commonly used mine machineries/equipments](#)
2. [MIN/N3212: Perform troubleshooting and repair of commonly used mine machineries/equipment](#)
3. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Electives (mandatory to select at least one):

Elective 1: Underground Metal

1. [MIN/N1702: Follow Health, Safety and Environmental guidelines for Underground Metalliferous Mines](#)

Elective 2: Opencast

1. [MIN/N1703: Follow Health, Safety, and Environmental Guidelines for opencast mines](#)

Elective 3: Underground Coal

1. [MIN/N1704: Follow Health, Safety, and Environmental guidelines for underground coal mines](#)

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Options(Not mandatory):

Option 1: Repair-Underground

1. [MIN/N3209: Troubleshooting and repair/ maintenance of UG mine equipment/machineries](#)

Option 2: Crusher

1. [MIN/N3210: Perform repair and maintenance of crusher and it's equipments](#)

Qualification Pack (QP) Parameters

Sector	Mining
Sub-Sector	Engineering Services
Occupation	Mechanical Services
Country	India
NSQF Level	3.0
Credits	21
Aligned to NCO/ISCO/ISIC Code	NCO-2015/7231.0401
Minimum Educational Qualification & Experience	10th grade pass (or equivalent) OR 9th grade pass (with 1.5 years' relevant experience in Mining Industry) OR 8th grade pass (with 3 years' relevant experience in Mining Industry) OR 5th grade pass (with 7.5 years' relevant experience in Mining Industry) OR Previous relevant Qualification of NSQF Level 2 (Helper-Opencast Mines or Helper-Underground Mines with 3 years relevant experience in Mining Industry)
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	Not Applicable



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Minimum Job Entry Age	20 Years
Last Reviewed On	NA
Next Review Date	06/05/2028
NSQC Approval Date	08/05/2025
Version	3.0
Reference code on NQR	QG-03-MG-04061-2025-V2-SCMS
NQR Version	1.0

Remarks:

Total Notional Duration: 390 Hours; Additional 60 hours for Optional NOS= Domain Theory Duration: 80 Hours; Additional 20 hours for Optional NOS Theory Duration+ Domain Practical Duration: 280 Hours; Additional 40 hours for Optional NOS Practical Duration+ Employability Skills Duration: 30 Hours

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MIN/N3211: Perform installation and preventive maintenance of commonly used mine machineries/equipments

Description

This unit is about installation and preventive maintenance activities for plant machinery, medium and light vehicles, pumps, compressors, pneumatic machines and other machine assemblies.

Scope

The scope covers the following :

- Installing machines, mechanical components and equipments
- Performing preventive maintenance of machine components in plant machinery, pumps, compressors, pneumatic and other machines/vehicles tracking
- Tracking and logging preventive maintenance, repairs, operational faults and other activities

Elements and Performance Criteria

Installing machines, mechanical components and equipments

To be competent, the user/individual on the job must be able to:

- PC1.** use ropes, slings, towing and lifting devices while assembling/disassembling machine/equipments and safely operate various types of hand and power tools
- PC2.** follow drawings and blue-prints given in the instruction sheet and installation manual
- PC3.** carry out various assembly of machines and conveyors etc.
- PC4.** follow the manufacturers instructions which apply to the care and safe handling of the machine/automobile
- PC5.** test assembled machine for proper performance before handing over for operations

Performing preventive maintenance of machine components in plant machinery, pumps, compressors, pneumatic and other machines/vehicles tracking

To be competent, the user/individual on the job must be able to:

- PC6.** follow maintenance schedule recommended by the equipment manufacturer
- PC7.** carry out greasing and lubrication of pivot points in a machine
- PC8.** open and re-assemble various types of bearings in machines
- PC9.** adjust valves, linkages, bearings, hydraulic, pneumatic and systems for smooth operation

Tracking and logging preventive maintenance, repairs, operational faults and other activities

To be competent, the user/individual on the job must be able to:

- PC10.** track hours-in-operation and adhere to preventive maintenance schedules of various machines assigned to them
- PC11.** maintain a checking/maintenance logbook to record all activities performed
- PC12.** inform the supervisor of problems that are beyond scope
- PC13.** ensure availability of fuel, lubricants, consumables and other supplies

Knowledge and Understanding (KU)

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The individual on the job needs to know and understand:

- KU1.** job-specific documents e.g. daily maintenance checklist and importance of the same
- KU2.** risk and impact of not following defined procedures/work instructions
- KU3.** the hierarchy for reporting identified problems
- KU4.** cost of equipment and loss for the company that results from damage of equipment
- KU5.** implications of delays in the process
- KU6.** handover and takeover procedures of the mine fitter according to company's SOP
- KU7.** safety guidelines specified by Directorate General of Mines Safety (DGMS) specific to mechanical operations
- KU8.** different types of mines and detail of the mine one is working in
- KU9.** benching in quarries, dressing of overhangs, undercuts, fencing
- KU10.** importance of first aid and hygiene
- KU11.** code of practice in specific areas of the mine
- KU12.** standing orders in force at the mine
- KU13.** importance of safety in the vicinity of machinery
- KU14.** about shot-firing / blasting related safety regulations including taking shelter during blasting
- KU15.** duties of workmen under the Mines Act-1952
- KU16.** provision of compensation and working hours, leaves, etc. as per Mines Act-1952
- KU17.** the outcome of violation of safety procedures
- KU18.** different types of machines used in open cast and underground mines and their specific functions
- KU19.** safety management plan prepared by the organization
- KU20.** various types of hand and power tools (spanners, jacks, drills etc.) and their use.
- KU21.** various types of lubricants and its importance, storage, handling etc.
- KU22.** various types of fasteners, nuts and bolts, threads, seals and couplings
- KU23.** various types of bearings used in machines and their assembly techniques
- KU24.** air systems, compressors its uses and pneumatic controls
- KU25.** various types of pumps and control valves of hydraulic systems
- KU26.** construction and operation crawlers
- KU27.** steering systems and various linkages
- KU28.** safety rules while using tools and tackling machine parts
- KU29.** different types of tyres and wheels used in medium and light vehicles, tracks, final drive etc., in case of crawling equipment
- KU30.** hot and cold tyre pressure as per size of tyre and methods of repairs
- KU31.** repair of overhauling electrical and mechanical engines, manual and power shift transmissions, hydraulic and pneumatic systems etc.
- KU32.** condition and performance of equipment using condition monitoring tools
- KU33.** Standard Operating Procedure (SOP) for performing preventive maintenance tasks
- KU34.** importance of sensitization towards different genders and PWD (Persons with Disabilities)

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Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** note down observations
- GS2.** read and interpret symbols and readings
- GS3.** read and write information documents or enter the information in register and entry of data in Management Information System (MIS)
- GS4.** read and interpret drawings, symbols, readings and measurements
- GS5.** read information documents
- GS6.** discuss task lists, schedules and activities
- GS7.** effectively communicate, listen and comprehend the information given by various sources about the site
- GS8.** operate computer and use its basic applications
- GS9.** understand and read basic english
- GS10.** make decisions pertaining to the concerned area of work
- GS11.** plan and organize the work order and tasks
- GS12.** organize all operation and service manuals so that sorting/ accessing information is easy
- GS13.** detect problems in day to day tasks
- GS14.** discuss possible solution with the supervisor for problem solving
- GS15.** make decisions in emergency conditions
- GS16.** follow instructions and work on areas of improvement identified
- GS17.** complete the assigned tasks timely
- GS18.** use reasoning skills to identify and resolve basic problems
- GS19.** analyze and detect any potential problems which could arise during operation

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Installing machines, mechanical components and equipments</i>	11	20	-	6
PC1. use ropes, slings, towing and lifting devices while assembling/disassembling machine/equipments and safely operate various types of hand and powertools	2	4	-	1
PC2. follow drawings and blue-prints given in the instruction sheet and installation manual	2	4	-	1
PC3. carry out various assembly of machines and conveyors etc.	2	4	-	1
PC4. follow the manufacturers instructions which apply to the care and safe handling of the machine/automobile	2	4	-	1
PC5. test assembled machine for proper performance before handing over for operations	3	4	-	2
<i>Performing preventive maintenance of machine components in plant machinery, pumps, compressors, pneumatic and other machines/vehicles tracking</i>	9	15	-	6
PC6. follow maintenance schedule recommended by the equipment manufacturer	2	4	-	2
PC7. carry out greasing and lubrication of pivot points in a machine	3	4	-	1
PC8. open and re-assemble various types of bearings in machines	2	3	-	2
PC9. adjust valves, linkages, bearings, hydraulic, pneumatic and systems for smooth operation	2	4	-	1
<i>Tracking and logging preventive maintenance, repairs, operational faults and other activities</i>	10	15	-	8
PC10. track hours-in-operation and adhere to preventive maintenance schedules of various machines assigned to them	2	4	-	2



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. maintain a checking/maintenance logbook to record all activities performed	2	3	-	2
PC12. inform the supervisor of problems that are beyond scope	3	4	-	2
PC13. ensure availability of fuel, lubricants, consumables and other supplies	3	4	-	2
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	MIN/N3211
NOS Name	Perform installation and preventive maintenance of commonly used mine machineries/equipments
Sector	Mining
Sub-Sector	Engineering Services
Occupation	Mechanical Services
NSQF Level	3
Credits	4
Version	2.0
Last Reviewed Date	08/05/2025
Next Review Date	06/05/2028
NSQC Clearance Date	08/05/2025

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MIN/N3212: Perform troubleshooting and repair of commonly used mine machineries/equipment

Description

This unit is about performing troubleshooting and repair activities of mechanical systems in plant and machinery, medium/light vehicles, compressors, pumps, pneumatic, hydraulic and other machines.

Scope

The scope covers the following :

- Performing troubleshooting and repair/maintenance
- Recording and Logging

Elements and Performance Criteria

Performing troubleshooting and repair/maintenance

To be competent, the user/individual on the job must be able to:

- PC1.** explain various elements of a good maintenance system and difference between preventive and repair maintenance
- PC2.** use various measuring instruments and testing tools
- PC3.** compare measured readings with optimal readings to pin-point faults
- PC4.** service, diagnose and repair faults in mechanical systems such as gears, steering systems, hydraulic pumps, transmission, crawlers, conveyor belts etc.
- PC5.** ensure the machine is on firm and level ground before attempting to carry out any maintenance activity
- PC6.** ensure the locking bar is in position to prevent the front and rear chassis moving and creating a crushing zone (articulated machines only)
- PC7.** ensure that no maintenance task on the machine/engine is performed when running or still hot
- PC8.** adjust valves, belt tensions for optimal operation
- PC9.** test repaired equipment to ensure everything is working correctly and safely (this may include Pre-load testing, parking brake testing, hydraulic pressure, greasing points, air pressure, etc.)

Recording and Logging

To be competent, the user/individual on the job must be able to:

- PC10.** complete daily/weekly maintenance/defect sheets as provided by the company
- PC11.** inform the supervisor of those problems which are not under one's purview
- PC12.** maintain inventory of fuel, lubricants and order other spares and consumables as required

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

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- KU1.** job-specific documents e.g. daily maintenance checklist and importance of the same
- KU2.** risk and impact of not following defined procedures/work instructions
- KU3.** the hierarchy for reporting identified problems
- KU4.** cost of equipment and loss for the company that results from damage of equipment
- KU5.** implications of delays in the process
- KU6.** handover and takeover procedures of the mine fitter according to company's SOP
- KU7.** safety guidelines specified by Directorate General of Mines Safety (DGMS) specific to mechanical operations
- KU8.** different types of mines and detail of the mine one is working in
- KU9.** benching in quarries, dressing of overhangs, undercuts, fencing
- KU10.** importance of first aid and hygiene
- KU11.** code of practice in specific areas of the mine
- KU12.** standing orders in force at the mine
- KU13.** importance of safety in the vicinity of machinery
- KU14.** about shot-firing/blasting related safety regulations including taking shelter during blasting
- KU15.** duties of workmen under the Mines Act-1952
- KU16.** provision of compensation and working hours, leaves, etc. as per Mines Act-1952
- KU17.** the outcome of violation of safety procedures
- KU18.** different types of machines used in open cast and underground mines and their specific functions
- KU19.** safety management plan prepared by the organization
- KU20.** basic calculations of volume, temperature, pressure, torque, unit conversions
- KU21.** various types of fasteners, nuts-bolts, threads, seals and couplings
- KU22.** various types of bearings used in machines and their assembly techniques
- KU23.** air systems, compressors, pneumatic controls and its uses
- KU24.** various types of pumps and control valves
- KU25.** construction and operation of crawlers
- KU26.** steering systems and various linkages
- KU27.** safety rules while using tools and tackling machine parts
- KU28.** repair of overhauling electrical and mechanical systems / engines, manual and power shift transmissions
- KU29.** condition and performance of equipment using condition monitoring tools
- KU30.** SOP for performing preventive maintenance jobs
- KU31.** importance of sensitization towards different genders and PWD (Persons with Disabilities)

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** note down observations
- GS2.** read and write information documents or enter the information in register and entry of data in Management Information System (MIS)



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- GS3.** read and interpret drawings, symbols, readings and measurements
- GS4.** discuss task lists, schedules and activities
- GS5.** effectively communicate, listen and comprehend the information given by various sources about the site
- GS6.** operate computer and use its basic applications
- GS7.** understand and read basic English
- GS8.** make decisions pertaining to the concerned area of work
- GS9.** plan and organize the work order and tasks
- GS10.** organize all operation and service manuals so that sorting/ accessing information is easy
- GS11.** detect problems in day to day tasks
- GS12.** discuss possible solution with the supervisor for problem solving
- GS13.** make decisions in emergency conditions
- GS14.** follow instructions and work on areas of improvement identified
- GS15.** complete the assigned tasks timely
- GS16.** use reasoning skills to identify and resolve basic problems
- GS17.** analyze and detect any potential problems which could arise during operation

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Performing troubleshooting and repair/maintenance</i>	24	38	-	16
PC1. explain various elements of a good maintenance system and difference between preventive and repair maintenance	3	4	-	2
PC2. use various measuring instruments and testing tools	2	4	-	2
PC3. compare measured readings with optimal readings to pin-point faults	3	4	-	2
PC4. service, diagnose and repair faults in mechanical systems such as gears, steering systems, hydraulic pumps, transmission, crawlers, conveyor belts etc.	3	4	-	2
PC5. ensure the machine is on firm and level ground before attempting to carry out any maintenance activity	3	4	-	1
PC6. ensure the locking bar is in position to prevent the front and rear chassis moving and creating a crushing zone (articulated machines only)	2	4	-	2
PC7. ensure that no maintenance task on the machine/engine is performed when running or still hot	3	5	-	2
PC8. adjust valves, belt tensions for optimal operation	2	4	-	1
PC9. test repaired equipment to ensure everything is working correctly and safely (this may include Pre-load testing, parking brake testing, hydraulic pressure, greasing points, air pressure, etc.)	3	5	-	2
<i>Recording and Logging</i>	6	12	-	4
PC10. complete daily/weekly maintenance/defect sheets as provided by the company	2	4	-	1



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. inform the supervisor of those problems which are not under one's purview	2	4	-	2
PC12. maintain inventory of fuel, lubricants and order other spares and consumables as required	2	4	-	1
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	MIN/N3212
NOS Name	Perform troubleshooting and repair of commonly used mine machineries/equipment
Sector	Mining
Sub-Sector	Engineering Services
Occupation	Mechanical Services
NSQF Level	3
Credits	6
Version	2.0
Last Reviewed Date	08/05/2025
Next Review Date	06/05/2028
NSQC Clearance Date	08/05/2025

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DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

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Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

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- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2029
NSQC Clearance Date	06/02/2026

Qualification Pack

MIN/N1702: Follow Health, Safety and Environmental guidelines for Underground Metalliferous Mines

Description

This unit is about adhering to health, safety and environmental guidelines in Underground Metalliferous Mines (UMM) during the execution of various tasks and operations and maintenance.

Scope

The scope covers the following :

- Follow work-site health and safety measures
- Follow environmental guidelines

Elements and Performance Criteria

Follow work-site health and safety measures

To be competent, the user/individual on the job must be able to:

- PC1.** undertake "The Take-5 (Personal Risk Assessment)" before commencement of any work
- PC2.** comply with safety, health and security-related regulations/guidelines at the mine e.g. follow Standard Operating Procedure (SOP) for material handling in underground (U/G) mine
- PC3.** operate various types of fire extinguishers to control different types of fire at a worksite when required
- PC4.** check that roof supporting is as per Systematic Support Plan (SSP) and approved Systematic Support Rules (SSR) while undertaking work in an area
- PC5.** take precaution against occupational health hazards (like dust, water, mine gases etc.) due to U/G working environment
- PC6.** use self-rescue apparatus, appropriately when required
- PC7.** follow appropriate emergency response procedure during emergency such as fire, water inrush, fall of ground etc.
- PC8.** follow precautions against U/G electrical appliances
- PC9.** follow appropriate Standard Operating Procedure while working near any isolated and sealed off area of the mine
- PC10.** follow appropriate safety practices while traveling on U/G haul roads, incase of post blast fumes and misfire
- PC11.** follow the manufacturer's instructions for care and safe operation of mine machinery and equipment
- PC12.** identify the working ventilation district from line diagram of ventilation circuit to direct air to the working face
- PC13.** follow Safety Management Plan (SMP) and Emergency Management Plan (EMP)
- PC14.** follow gas detecting alarm signal on leakage of inflammable gases and laid out procedure to ensure safety
- PC15.** follow process for reporting any unsafe act/condition in work area to the concerned person

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- PC16.** use underground mine communication system
- PC17.** ensure positive isolation near the work place if applicable
- PC18.** use appropriate PPE as per the requirement
- PC19.** maintain hand hygiene by washing hands with alcohol based sanitisers/soap
- PC20.** disinfect the machine/tools before and after work/task
- PC21.** maintain hygiene at the work site
- PC22.** report any symptoms of illness to the shift-incharge
- PC23.** identify six directional hazards at workplace and take decisions accordingly

Follow environmental guidelines

To be competent, the user/individual on the job must be able to:

- PC24.** identify the environmental impact of mining related operations and follow steps to reduce those impact
- PC25.** follow the mineral conservation practices in U/G mining operations to achieve optimum ore or mineral recovery
- PC26.** ensure that the stowing practices produce minimum disturbance to the surface
- PC27.** ensure that the subgrade ore is carried out to surface and stacked separately at the earmarked place
- PC28.** ensure the productivity of the machine for material/fuel conservation
- PC29.** follow the process for collecting, storing and disposing of the hazardous material and waste (like used oil, lubricant, battery, etc.) in compliance with worksite guidelines
- PC30.** follow the "5-S" practice at work site like cleaning oil from ground (to avoid soil from getting damaged), etc.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** safety guidelines specified by Directorate General of Mine Safety (DGMS)
- KU2.** duties and rights of workers
- KU3.** selection process of person for rescue training
- KU4.** about Systematic Support Plan (SSP) and approved Systematic Support Rules (SSR)
- KU5.** isolation and sealed off area of the mine
- KU6.** various types of gases found in the mine and their effect
- KU7.** self-rescue apparatus and their uses
- KU8.** provision of medical examination (IME & PME) of person employed as per Mines Rules 1955
- KU9.** importance of first aid and hygiene
- KU10.** importance of blind spot survey/training
- KU11.** about different types of machinery used in U/G mines
- KU12.** different types of supporting systems used in U/G mines as per SSP and SSR
- KU13.** about precautions to be taken when handling heavy equipment
- KU14.** various problems/incidents likely to occur
- KU15.** role of Internal Safety Organization, safety committee, workman's inspector and DGMS

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- KU16.** about mine safety standard including light illumination level, noise levels, dust level, pollutants, etc at the work-site
- KU17.** common sources of pollution in the mines and ways to minimize it
- KU18.** various types of fire extinguishers
- KU19.** safety equipment like safety shoes, safety belt, tight fit clothing, hand gloves, safety goggles, Gas Detector, Safety Lamp, Self-Contained Breathing Apparatus, gum boots, ear plugs, Face Mask, etc.
- KU20.** shot-firing / blasting related safety regulations including taking shelter during blasting
- KU21.** emergency response /disaster management plan prepared by the organization as per DGMS guideline
- KU22.** mining area-specific signs, and other safety and emergency signals
- KU23.** the outcome of violation of safety procedures
- KU24.** Take-5 (Personal Risk Assessment) training (DGMS Tech. circulars 2/2014)
- KU25.** rules and regulations for safety and security while handling hazardous materials
- KU26.** safety appliances and rescue equipment
- KU27.** importance and use of various communication system used in UG mines
- KU28.** importance of positive isolation at working site
- KU29.** safety and occupational health policy of organisation
- KU30.** six directional hazard identification process
- KU31.** basic personal and workplace hygiene
- KU32.** importance of FAB (Fresh Air Base)
- KU33.** basic provisions in Mines Creche Rules, 1966 (MCR) for females employed in the mines
- KU34.** the role and responsibilities of rescue room and rescue station and how to contact them in case of emergency
- KU35.** importance of taking shelter at the miner's station during blasting operation
- KU36.** importance of sensitization towards different genders and persons with disabilities (PWD)
- KU37.** importance of following infection control policies, '5-S' practices, and waste management
- KU38.** importance of water/material/energy conservation and management
- KU39.** importance of SMP and EMP, prepared by the organization, as per DGMS guideline

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** fill up documentation applicable to one's role
- GS2.** make decisions on the concerned area of work
- GS3.** read basic English language
- GS4.** read and interpret manuals, health, and safety instructions, memos, etc.
- GS5.** use the digital information from machine
- GS6.** use basic applications of a computer
- GS7.** plan and organize the work order and tasks
- GS8.** use reasoning skills to identify and resolve fundamental problems

Qualification Pack

GS9. complete the assigned tasks timely

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow work-site health and safety measures</i>	24	42	-	17
PC1. undertake "The Take-5 (Personal Risk Assessment)" before commencement of any work	-	2	-	2
PC2. comply with safety, health and security-related regulations/guidelines at the mine e.g. follow Standard Operating Procedure (SOP) for material handling in underground (U/G) mine	1	3	-	-
PC3. operate various types of fire extinguishers to control different types of fire at a worksite when required	1	4	-	1
PC4. check that roof supporting is as per Systematic Support Plan (SSP) and approved Systematic Support Rules (SSR) while undertaking work in an area	1	1	-	-
PC5. take precaution against occupational health hazards (like dust, water, mine gases etc.) due to U/G working environment	1	2	-	1
PC6. use self-rescue apparatus, appropriately when required	-	4	-	1
PC7. follow appropriate emergency response procedure during emergency such as fire, water inrush, fall of ground etc.	1	2	-	1
PC8. follow precautions against U/G electrical appliances	2	2	-	2
PC9. follow appropriate Standard Operating Procedure while working near any isolated and sealed off area of the mine	2	2	-	1
PC10. follow appropriate safety practices while traveling on U/G haul roads, incase of post blast fumes and misfire	2	4	-	-
PC11. follow the manufacturer's instructions for care and safe operation of mine machinery and equipment	1	2	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. identify the working ventilation district from line diagram of ventilation circuit to direct air to the working face	1	1	-	-
PC13. follow Safety Management Plan (SMP) and Emergency Management Plan (EMP)	1	-	-	-
PC14. follow gas detecting alarm signal on leakage of inflammable gases and laid out procedure to ensure safety	1	2	-	1
PC15. follow process for reporting any unsafe act/condition in work area to the concerned person	-	2	-	1
PC16. use underground mine communication system	-	1	-	-
PC17. ensure positive isolation near the work place if applicable	1	1	-	1
PC18. use appropriate PPE as per the requirement	2	4	-	2
PC19. maintain hand hygiene by washing hands with alcohol based sanitisers/soap	1	1	-	1
PC20. disinfect the machine/tools before and after work/task	1	1	-	-
PC21. maintain hygiene at the work site	1	1	-	1
PC22. report any symptoms of illness to the shift-incharge	1	-	-	-
PC23. identify six directional hazards at workplace and take decisions accordingly	2	-	-	-
<i>Follow environmental guidelines</i>	6	8	-	3
PC24. identify the environmental impact of mining related operations and follow steps to reduce those impact	1	2	-	-
PC25. follow the mineral conservation practices in U/G mining operations to achieve optimum ore or mineral recovery	-	1	-	-
PC26. ensure that the stowing practices produce minimum disturbance to the surface	1	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC27. ensure that the subgrade ore is carried out to surface and stacked separately at the earmarked place	1	-	-	1
PC28. ensure the productivity of the machine for material/fuel conservation	1	1	-	-
PC29. follow the process for collecting, storing and disposing of the hazardous material and waste (like used oil, lubricant, battery, etc.) in compliance with worksite guidelines	1	1	-	1
PC30. follow the "5-S" practice at work site like cleaning oil from ground (to avoid soil from getting damaged), etc.	1	2	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1702
NOS Name	Follow Health, Safety and Environmental guidelines for Underground Metalliferous Mines
Sector	Mining
Sub-Sector	Mining Operation
Occupation	HSE Functions, Mine Surveying, Drilling/Cutting, Shot firing/Blasting, Loading and Hauling - Underground, Specialized Operations, Electrical Services, Mechanical Services
NSQF Level	1
Credits	2
Version	3.0
Last Reviewed Date	08/05/2025
Next Review Date	06/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

MIN/N1703: Follow Health, Safety, and Environmental Guidelines for opencast mines

Description

This unit is about adhering to health, safety and environmental guidelines at the Open Cast Mines while executing various tasks, maintenance and operations.

Scope

The scope covers the following :

- Follow work-site health and safety measures
- Follow Environmental guidelines

Elements and Performance Criteria

Follow work-site health and safety measures

To be competent, the user/individual on the job must be able to:

- PC1.** comply with safety, health, and security-related regulations/guidelines at the opencast mine
- PC2.** follow the safety instructions given by the workman's inspector
- PC3.** follow adequate safety while working at haul roads, heights, overburden dumps, sump area, stockyard, near moving parts, etc.
- PC4.** take safety precautions while working on sites (sub-station, workshop etc.), with equipment, and conducting welding and cutting operations
- PC5.** follow appropriate Safe Operating Procedure (SOP) while dealing with explosives
- PC6.** respond promptly and appropriately to an accident/ incident or an emergency situation, within limits of the role and responsibility
- PC7.** provide first aid to an injured person
- PC8.** operate various types of fire extinguishers to control different types of fire at a worksite when required
- PC9.** use appropriate PPE as per the requirement
- PC10.** maintain hand hygiene by washing hands with alcohol based sanitisers/soap
- PC11.** disinfect the machine/tools before and after work/task
- PC12.** maintain hygiene at the work site
- PC13.** report any symptoms of illness to the shift-incharge

Follow Environmental guidelines

To be competent, the user/individual on the job must be able to:

- PC14.** identify the environmental impact of related opencast mining operations
- PC15.** follow the process for collecting, storing and disposing of the hazardous material and waste (like used oil, lubricant, battery, etc.) in compliance with worksite guidelines
- PC16.** ensure not to mix topsoil with waste in day to day tasks
- PC17.** ensure that HEMM is washed at the designated location

Qualification Pack

- PC18.** ensure the productivity of the machine for material/fuel conservation
- PC19.** follow the mineral conservation practices specified by the organization in accordance with MCDR-2017 (Mineral Conservation and Development Rules)
- PC20.** assist supervisor for reducing environmental impact caused due to related mining operations

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** about various environmental awareness program related to mining, organized by the various government bodies/company
- KU2.** safety guidelines specified by Directorate General of Mine Safety (DGMS)
- KU3.** how to identify six-directional hazards
- KU4.** basic mining terminologies and definitions
- KU5.** importance of blind spot survey/training
- KU6.** means of access and egress from the mines, location of workshop, haul roads and working face including dump yards
- KU7.** duties of workers under The Mines act-1952
- KU8.** working hours and accident compensation as per The Mines act-1952
- KU9.** about precautions to be taken when handling heavy equipment
- KU10.** various problems/incidents likely to occur
- KU11.** hierarchy of the reporting
- KU12.** machine operation, condition of the machine and worksite
- KU13.** proper documents specific to the machine
- KU14.** role of workmen inspector, safety committee and internal safety organization
- KU15.** the process of top soil removal and management
- KU16.** mine sump and pumping system of the mines
- KU17.** about mine safety standard including light illumination level, noise levels, dust level, pollutants, etc at the work-site
- KU18.** common sources of pollution in the mines and ways to minimize it
- KU19.** safety equipment like safety shoes, safety belt, tight fit clothing, hand gloves, safety goggles, gas detector, safety lamp, self-contained breathing apparatus, gum boots, ear plugs, face mask, etc.
- KU20.** shot-firing / blasting related safety regulations including taking shelter during blasting
- KU21.** emergency response /disaster management plan prepared by the organization
- KU22.** signages, mining area-specific signs, and other safety and emergency signals
- KU23.** the outcome of violation of safety procedures
- KU24.** basic personal and workplace hygiene
- KU25.** importance of sensitization towards different genders and PWD (Persons with Disabilities)

Generic Skills (GS)

User/individual on the job needs to know how to:



Skill Council For Mining Sector



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- GS1.** fill up documentation applicable to one's role
- GS2.** make decisions on the concerned area of work
- GS3.** read and interpret manuals, health, and safety instructions, memos, etc.
- GS4.** plan and organize the work order and jobs
- GS5.** use reasoning skills to identify and resolve fundamental problems
- GS6.** complete the assigned tasks timely

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow work-site health and safety measures</i>	20	34	-	15
PC1. comply with safety, health, and security-related regulations/guidelines at the opencast mine	2	2	-	1
PC2. follow the safety instructions given by the workman's inspector	1	3	-	1
PC3. follow adequate safety while working at haul roads, heights, overburden dumps, sump area, stockyard, near moving parts, etc.	3	3	-	1
PC4. take safety precautions while working on sites (sub-station, workshop etc.), with equipment, and conducting welding and cutting operations	1	3	-	1
PC5. follow appropriate Safe Operating Procedure (SOP) while dealing with explosives	2	3	-	2
PC6. respond promptly and appropriately to an accident/ incident or an emergency situation, within limits of the role and responsibility	2	3	-	1
PC7. provide first aid to an injured person	2	3	-	1
PC8. operate various types of fire extinguishers to control different types of fire at a worksite when required	1	3	-	1
PC9. use appropriate PPE as per the requirement	2	4	-	2
PC10. maintain hand hygiene by washing hands with alcohol based sanitisers/soap	1	2	-	1
PC11. disinfect the machine/tools before and after work/task	1	2	-	1
PC12. maintain hygiene at the work site	1	1	-	1
PC13. report any symptoms of illness to the shift-incharge	1	2	-	1
<i>Follow Environmental guidelines</i>	10	16	-	5

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. identify the environmental impact of related opencast mining operations	2	2	-	1
PC15. follow the process for collecting, storing and disposing of the hazardous material and waste (like used oil, lubricant, battery, etc.) in compliance with worksite guidelines	1	2	-	1
PC16. ensure not to mix topsoil with waste in day to day tasks	2	2	-	1
PC17. ensure that HEMM is washed at the designated location	2	2	-	1
PC18. ensure the productivity of the machine for material/fuel conservation	1	3	-	-
PC19. follow the mineral conservation practices specified by the organization in accordance with MCDR-2017 (Mineral Conservation and Development Rules)	1	3	-	1
PC20. assist supervisor for reducing environmental impact caused due to related mining operations	1	2	-	-
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1703
NOS Name	Follow Health, Safety, and Environmental Guidelines for opencast mines
Sector	Mining
Sub-Sector	Mining Operation
Occupation	HSE Functions, , Exploration, Mineral Estimation, Planning, Mine Surveying, Drilling/Cutting, Shot firing/Blasting, Loading and Hauling - Opencast, Specialized Operations, Electrical Services, Mechanical Services, Instrumentation and Control Systems, Ore Processing
NSQF Level	1
Credits	2
Version	3.0
Last Reviewed Date	08/05/2025
Next Review Date	06/05/2028
NSQF Clearance Date	08/05/2025

Qualification Pack

MIN/N1704: Follow Health, Safety, and Environmental guidelines for underground coal mines

Description

This unit is about adhering to health, safety and environmental guidelines in Underground Coal Mines (UCM) during execution of various tasks, operations and maintenance.

Scope

The scope covers the following :

- Follow work-site health and safety measures
- Follow environmental guidelines

Elements and Performance Criteria

Follow work-site health and safety measures

To be competent, the user/individual on the job must be able to:

- PC1.** follow preventive measures against firedamp, whitedamp, blackdamp etc.
- PC2.** use the flame safety lamp for detecting the methane gas as per Standard Operating Procedure (SOP)
- PC3.** undertake "The Take-5 (Personal Risk Assessment)" before commencement of any work
- PC4.** comply with safety, health and security-related regulations/guidelines at the mine e.g. SOP for material handling in underground (U/G) mine
- PC5.** ensure that oil, grease, canvas or other inflammable material are stored in fire-proof receptacle
- PC6.** ensure that every instrument, apparatus and equipment are DGMS approved before these are used
- PC7.** ensure that Armoured face conveyor (AFC) and chocks must be kept in a straight line for every cycle of operations and tightened up to the setting pressure while keeping it in full contact with the roof, applicable for longwall mining
- PC8.** provide first aid to an injured person
- PC9.** follow safety precautions against spontaneous heating of the coal
- PC10.** operate various types of fire extinguishers to control different types of fire at worksite, if required
- PC11.** ensure that no person is traveling/working/staying under unsupported roof
- PC12.** check that roof supporting is as per Systematic Support Plan (SSP) and approved Systematic Support Rules (SSR) while undertaking work in an area
- PC13.** take precaution against occupational health hazards (like dust, water, mine gases etc.) due to U/G working environment
- PC14.** use self-rescue apparatus appropriately when required
- PC15.** follow Safety Management Plan (SMP) and Emergency Management Plan (EMP)
- PC16.** follow precautions against U/G electrical appliances

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- PC17.** take proper care against damage and accidents while loading, transporting, dismantling and erecting of roof supports
- PC18.** follow appropriate SOP while working near any isolated and sealed off area of the mine
- PC19.** ensure that the roof and the sidewalls of the mine face (or newly exposed area of the mines) have been scaled/dressed properly
- PC20.** take relevant safety precautions during depillaring operation in UCM
- PC21.** follow appropriate safety practices while traveling on U/G haul roads, incase of post blast fumes and misfire
- PC22.** follow the manufacturer's instructions for care and safe operation of mine machinery and equipment
- PC23.** identify the working ventilation district from line diagram of ventilation circuit to direct air to the working face
- PC24.** follow laid out SOP in case of alarm signal for leakage of inflammable gases
- PC25.** follow the process of reporting any unsafe act/condition in the working area to the concerned person
- PC26.** use underground mine communication system
- PC27.** ensure positive isolation near the work place if applicable
- PC28.** use appropriate Personal Protective Equipment (PPE) as per the requirement
- PC29.** maintain hand hygiene by washing hands with alcohol based sanitisers/soap
- PC30.** disinfect the machine/tools before and after work/task
- PC31.** maintain hygiene at the work site
- PC32.** report any symptoms of illness to the shift-incharge
- PC33.** identify six directional hazards at workplace and take decisions accordingly

Follow environmental guidelines

To be competent, the user/individual on the job must be able to:

- PC34.** identify the environmental impact of mining related operations and follow steps to reduce those impact
- PC35.** follow the mineral conservation practices in U/G mining operations to achieve optimum ore or mineral recovery
- PC36.** ensure that the stowing practices produce minimum disturbance to the surface
- PC37.** ensure that the subgrade coal is carried out to surface and stacked separately at the earmarked place
- PC38.** ensure the productivity of the machine for material/fuel conservation
- PC39.** follow the process for collecting, storing and disposing of the hazardous material and waste (like used oil, lubricant, battery, etc.) in compliance with worksite guidelines
- PC40.** follow the "5-S" practice at work site like cleaning oil from ground (to avoid soil from getting damaged), etc.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** safety guidelines specified by Directorate General of Mine Safety (DGMS)

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- KU2.** duties and rights of workers
- KU3.** selection process of person for rescue training
- KU4.** about Systematic Support Plan (SSP) and approved Systematic Support Rules (SSR)
- KU5.** isolation and sealed off area of the mine
- KU6.** various types of gases available in the mine and their effects; and their control measures
- KU7.** self-rescue apparatus and their uses
- KU8.** provision of medical examination (Initial Medical Examination (IME) & Periodical Medical Examination (PME)) of a person employed, as per Mines Rules 1955
- KU9.** importance of first aid and hygiene
- KU10.** about different types of machineries used in U/G mines
- KU11.** different types of supporting system used in U/G mines as per SSP and SSR
- KU12.** about precautions to be taken when handling heavy equipment
- KU13.** various problems/incidents likely to occur
- KU14.** role of Internal Safety Organization, safety committee, workman's inspector and DGMS
- KU15.** about mine safety standard including light illumination level, noise levels, dust level, pollutants, etc at the work-site
- KU16.** common sources of pollution in the mines and ways to minimize it
- KU17.** various types of fire extinguishers
- KU18.** safety equipment like safety shoes, safety belt, tight fit clothing, hand gloves, safety goggles, Gas Detector, Safety Lamp, Self-Contained Breathing Apparatus, gum boots, ear plugs, Face Mask, etc.
- KU19.** shot-firing / blasting related safety regulations including taking shelter during blasting
- KU20.** mining area-specific signs, and other safety and emergency signals
- KU21.** the outcome of violation of safety procedures
- KU22.** Take-5 (Personal Risk Assessment) training (DGMS Tech. circulars 2/2014)
- KU23.** hazardous material safety, security rules and regulations
- KU24.** safety appliances and rescue equipment
- KU25.** importance and use of various communication system used in UG mines
- KU26.** importance of positive isolation at working site
- KU27.** safety and occupational health policy of organisation
- KU28.** six directional hazard identification process
- KU29.** importance of blind spot survey/training
- KU30.** basic personal and workplace hygiene
- KU31.** importance of FAB (Fresh Air Base)
- KU32.** basic provisions in Mines Creche Rules, 1966 (MCR) for any females employed in the mines
- KU33.** about basic safety regulations of Coal Mines Regulation, 2017 (CMR)
- KU34.** types of stone dust barrier and its importance
- KU35.** coal dust explosion and its preventive measures
- KU36.** classification of coal mines as per the degree of gassiness of coal seams such as first degree, second degree, and third-degree mines
- KU37.** precautions as per the gassiness of the coal mines

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- KU38.** use of flame safety lamp and its parts
- KU39.** about coal mines occupational disease such as pneumoconiosis or 'black lung' and their preventive measures
- KU40.** Standard of Ventilation as per the section 153 of the CMR 2017
- KU41.** Standard of Lighting as per the section 178 of the CMR 2017
- KU42.** the roles, duties and responsibilities of rescue team members, rescue room and rescue station and how to contact them in case of emergency
- KU43.** the correct steps for conducting any rescue work as per Mine Rescue Rule (MRR)
- KU44.** importance of taking shelter at the miner's station during blasting operation
- KU45.** importance of sensitization towards different genders and persons with disabilities (PWD)
- KU46.** importance of following infection control policies, '5-S' practices, and waste management
- KU47.** importance of water/material/energy conservation and management
- KU48.** importance of SMP and EMP, prepared by the organization, as per DGMS guideline

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** fill up documentation applicable to one's role
- GS2.** make decisions on the concerned area of work
- GS3.** read basic English language
- GS4.** read and interpret manuals, health, and safety instructions, memos, etc.
- GS5.** use the digital information from a machine
- GS6.** plan and organize the work order and tasks
- GS7.** use basic applications of the computer
- GS8.** use reasoning skills to identify and resolve fundamental problems
- GS9.** complete the assigned tasks timely

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow work-site health and safety measures</i>	24	43	-	14
PC1. follow preventive measures against firedamp, whitedamp, blackdamp etc.	1	1	-	-
PC2. use the flame safety lamp for detecting the methane gas as per Standard Operating Procedure (SOP)	-	1	-	-
PC3. undertake "The Take-5 (Personal Risk Assessment)" before commencement of any work	1	1	-	1
PC4. comply with safety, health and security-related regulations/guidelines at the mine e.g. SOP for material handling in underground (U/G) mine	1	1	-	1
PC5. ensure that oil, grease, canvas or other inflammable material are stored in fire-proof receptacle	-	1	-	-
PC6. ensure that every instrument, apparatus and equipment are DGMS approved before these are used	1	1	-	-
PC7. ensure that Armoured face conveyor (AFC) and chocks must be kept in a straight line for every cycle of operations and tightened up to the setting pressure while keeping it in full contact with the roof, applicable for longwall mining	3	3	-	-
PC8. provide first aid to an injured person	1	2	-	1
PC9. follow safety precautions against spontaneous heating of the coal	1	1	-	-
PC10. operate various types of fire extinguishers to control different types of fire at worksite, if required	-	2	-	-
PC11. ensure that no person is traveling/working/staying under unsupported roof	1	-	-	-
PC12. check that roof supporting is as per Systematic Support Plan (SSP) and approved Systematic Support Rules (SSR) while undertaking work in an area	-	2	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. take precaution against occupational health hazards (like dust, water, mine gases etc.) due to U/G working environment	-	1	-	1
PC14. use self-rescue apparatus appropriately when required	-	2	-	-
PC15. follow Safety Management Plan (SMP) and Emergency Management Plan (EMP)	1	2	-	1
PC16. follow precautions against U/G electrical appliances	2	2	-	1
PC17. take proper care against damage and accidents while loading, transporting, dismantling and erecting of roof supports	1	2	-	-
PC18. follow appropriate SOP while working near any isolated and sealed off area of the mine	1	1	-	1
PC19. ensure that the roof and the sidewalls of the mine face (or newly exposed area of the mines) have been scaled/dressed properly	1	1	-	-
PC20. take relevant safety precautions during depillaring operation in UCM	-	2	-	1
PC21. follow appropriate safety practices while traveling on U/G haul roads, incase of post blast fumes and misfire	1	1	-	-
PC22. follow the manufacturer's instructions for care and safe operation of mine machinery and equipment	-	1	-	1
PC23. identify the working ventilation district from line diagram of ventilation circuit to direct air to the working face	-	1	-	1
PC24. follow laid out SOP in case of alarm signal for leakage of inflammable gases	1	1	-	-
PC25. follow the process of reporting any unsafe act/condition in the working area to the concerned person	-	1	-	1
PC26. use underground mine communication system	-	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC27. ensure positive isolation near the work place if applicable	1	1	-	-
PC28. use appropriate Personal Protective Equipment (PPE) as per the requirement	2	2	-	1
PC29. maintain hand hygiene by washing hands with alcohol based sanitisers/soap	-	1	-	-
PC30. disinfect the machine/tools before and after work/task	1	1	-	-
PC31. maintain hygiene at the work site	1	1	-	-
PC32. report any symptoms of illness to the shift-incharge	1	1	-	-
PC33. identify six directional hazards at workplace and take decisions accordingly	-	1	-	1
<i>Follow environmental guidelines</i>	6	7	-	6
PC34. identify the environmental impact of mining related operations and follow steps to reduce those impact	1	1	-	1
PC35. follow the mineral conservation practices in U/G mining operations to achieve optimum ore or mineral recovery	1	1	-	1
PC36. ensure that the stowing practices produce minimum disturbance to the surface	1	1	-	-
PC37. ensure that the subgrade coal is carried out to surface and stacked separately at the earmarked place	1	1	-	1
PC38. ensure the productivity of the machine for material/fuel conservation	1	1	-	1
PC39. follow the process for collecting, storing and disposing of the hazardous material and waste (like used oil, lubricant, battery, etc.) in compliance with worksite guidelines	-	1	-	1
PC40. follow the "5-S" practice at work site like cleaning oil from ground (to avoid soil from getting damaged), etc.	1	1	-	1



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1704
NOS Name	Follow Health, Safety, and Environmental guidelines for underground coal mines
Sector	Mining
Sub-Sector	Mining Operation
Occupation	HSE Functions, Mine Surveying, Drilling/Cutting, Shot firing/Blasting, Loading and Hauling - Underground, Specialized Operations, Electrical Services, Mechanical Services
NSQF Level	1
Credits	2
Version	3.0
Last Reviewed Date	08/05/2025
Next Review Date	06/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

MIN/N3209: Troubleshooting and repair/ maintenance of UG mine equipment/machineries

Description

This OS unit is about troubleshooting and repair/ maintenance of UG mine equipments/machineries such as ventilation fan, booster fan, power support, winding drum, skip system, haulage system, chute etc.

Scope

The scope covers the following :

- Perform troubleshooting and repair/ maintenance of UG mine equipments/machineries

Elements and Performance Criteria

Perform troubleshooting and repair/ maintenance of UG mine equipment/machineries

To be competent, the user/individual on the job must be able to:

- PC1.** conduct maintenance of the compressor of UG mine equipment, so that it operates properly and generates required output
- PC2.** check and perform basic maintenance and troubleshooting of winding engine drum and its various equipments such as rope joint, slings, cage, skip etc.
- PC3.** check and perform maintenance of ventilation fan, booster fan and its various sub-parts
- PC4.** identify and repair/reduce the air pressure loss in the ventilation system
- PC5.** check and repair any fault in the hoisting or winding system such as head-frame or head-gear, rope equipments and their various safety devices
- PC6.** perform maintenance/repair of the track and mine car if required
- PC7.** conduct the repairs/install of the various support system used in the mines such as hydraulic power support, rope support, steel props, shield support, rock bolting and its supporting equipments
- PC8.** perform various repairs of the haulage system such as various rope haulage, rail and track mounted rope haulage and other fittings etc.
- PC9.** ensure proper working of all safety systems used in the mines
- PC10.** repair the various pumps and their supporting equipment/structure

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** job-specific documents e.g. daily maintenance checklist and importance of the same
- KU2.** risk and impact of not following defined procedures/work instructions
- KU3.** the hierarchy for reporting identified problems
- KU4.** cost of equipment and loss for the company that results from damage of equipment
- KU5.** implications of delays in the process

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- KU6.** handover and takeover procedures of the mine fitter/mechanic according to company's SOP
- KU7.** safety guidelines specified by Directorate General of Mines Safety (DGMS) specific to various equipments used in the mines
- KU8.** different types of mines and detail of the mine one is working in
- KU9.** importance of first aid and hygiene
- KU10.** code of practice in specific areas of the mine
- KU11.** standing orders in force at the mine
- KU12.** importance of safety in the vicinity of machinery
- KU13.** about shot-firing/blasting related safety regulations including taking shelter during blasting
- KU14.** duties of workmen under the Mines Act-1952
- KU15.** provision of compensation and working hours, leaves, etc. as per Mines Act-1952
- KU16.** the outcome of violation of safety procedures
- KU17.** use of the compressor in mines and its various parameters for required output
- KU18.** various types of winding engine drum/system and its various equipments such as rope joint, slings, cage, skip etc.
- KU19.** importance of ventilation fan, booster fan and its various sub-parts and its impact by changing various parameters
- KU20.** standard mine ventilation parameter as per the DGMS
- KU21.** various types of the hoisting or winding system such as head-frame or head-gear, rope equipments and their various safety devices and their uses
- KU22.** use of various tools for repairing/installing the track and mine car and its applications
- KU23.** various support system used in the mines such as hydraulic power support, rope support, steel props, shield support, rock bolting and its supporting equipment and their importances
- KU24.** various haulage system such as various rope haulage, rail and track mounted rope haulage, and other fittings etc.
- KU25.** various types of pump used for dewatering and its supporting equipments

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** note down observations (if any)
- GS2.** read and interpret symbols and readings
- GS3.** read information documents
- GS4.** discuss task lists, schedules and activities
- GS5.** effectively communicate, listen and comprehend the information given by various sources about the site
- GS6.** make decisions pertaining to the concerned area of work
- GS7.** plan and organize the work order and tasks
- GS8.** organize all operation and service manuals so that sorting/ accessing information is easy
- GS9.** detect problems in day to day tasks
- GS10.** discuss possible solution with the supervisor for problem solving



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Qualification Pack

- GS11.** make decisions in emergency conditions
- GS12.** follow instructions and work on areas of improvement identified
- GS13.** complete the assigned tasks timely
- GS14.** use reasoning skills to identify and resolve basic problems
- GS15.** analyze and detect any potential problems which could arise during operation
- GS16.** respect persons with disabilities
- GS17.** be sensitive towards all genders

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Perform troubleshooting and repair/ maintenance of UG mine equipment/machineries</i>	30	50	-	20
PC1. conduct maintenance of the compressor of UG mine equipment, so that it operates properly and generates required output	3	5	-	2
PC2. check and perform basic maintenance and troubleshooting of winding engine drum and its various equipments such as rope joint, slings, cage, skip etc.	3	5	-	2
PC3. check and perform maintenance of ventilation fan, booster fan and its various sub-parts	3	5	-	2
PC4. identify and repair/reduce the air pressure loss in the ventilation system	3	5	-	2
PC5. check and repair any fault in the hoisting or winding system such as head-frame or head-gear, rope equipments and their various safety devices	3	5	-	2
PC6. perform maintenance/repair of the track and mine car if required	3	5	-	2
PC7. conduct the repairs/install of the various support system used in the mines such as hydraulic power support, rope support, steel props, shield support, rock bolting and its supporting equipments	3	5	-	2
PC8. perform various repairs of the haulage system such as various rope haulage, rail and track mounted rope haulage and other fittings etc.	3	5	-	2
PC9. ensure proper working of all safety systems used in the mines	3	5	-	2
PC10. repair the various pumps and their supporting equipment/structure	3	5	-	2
NOS Total	30	50	-	20



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N3209
NOS Name	Troubleshooting and repair/ maintenance of UG mine equipment/machineries
Sector	Mining
Sub-Sector	Engineering Services
Occupation	Mechanical Services
NSQF Level	3
Credits	2
Version	2.0
Last Reviewed Date	08/05/2025
Next Review Date	06/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

MIN/N3210: Perform repair and maintenance of crusher and it's equipments

Description

This OS unit is about repair and maintenance of various types of crushers and it's various crushing equipments such as conveyor, chute, etc.

Scope

The scope covers the following :

- Perform repair and maintenance of various types of crushers and it's equipments

Elements and Performance Criteria

Perform repair and maintenance of various types of crushers and it's equipments

To be competent, the user/individual on the job must be able to:

- PC1.** discuss various types of crusher and their crushing principle
- PC2.** identify the important components of a jaw crusher, cone crusher, etc
- PC3.** check different types of crushing principal components such as gates, chutes and gaps, feeder, conveyor, dust collectors, water sprays etc.
- PC4.** conduct maintenance of the crushing plant
- PC5.** perform daily checking of plant as per check-list followed by organization
- PC6.** change stoke and re- assembly properly
- PC7.** check and replace thrust plate of the crusher
- PC8.** perform replacement of various worn out parts of the crusher
- PC9.** replace main shaft protection sleeve of the crusher
- PC10.** install upper frame assembly of various crusher
- PC11.** replace various worn out parts of vibrating screen, feeder, conveyor etc.
- PC12.** test samples, materials or products to ensure compliance with Indian standard (IS) specifications
- PC13.** conduct operational checks on areas of potential issues like unusual noises/smells, blockages and obstruction, leaks etc. and take corrective actions if required
- PC14.** check indicators that signal need for replacement
- PC15.** replenish coolants, lubricants, fluids and screeners as and when required
- PC16.** follow the safe code of practice for erection, installation, operation, repairs, maintenance, dismantling of plant and ancillary equipments
- PC17.** identify missing or defective components or controls and replace them with genuine OEM recommended components
- PC18.** check oil, fuel tanks for leaks and take necessary actions as per the operational manual
- PC19.** lubricate all the moving parts at regular intervals

Qualification Pack

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** job-specific documents e.g. daily maintenance checklist and importance of the same
- KU2.** risk and impact of not following defined procedures/work instructions
- KU3.** the hierarchy for reporting identified problems
- KU4.** cost of equipment and loss for the company that results from damage of equipment
- KU5.** implications of delays in the process
- KU6.** handover and takeover procedures of the mine fitter/mechanic according to company's SOP
- KU7.** safety guidelines specified by Directorate General of Mines Safety (DGMS) specific to various equipment used in the mines
- KU8.** different types of mines and detail of the mine one is working in
- KU9.** importance of first aid and hygiene
- KU10.** code of practice in specific areas of the mine
- KU11.** standing orders in force at the mine
- KU12.** importance of safety in the vicinity of machinery
- KU13.** about shot-firing / blasting related safety regulations including taking shelter during blasting
- KU14.** duties of workmen under the Mines Act-1952
- KU15.** provision of compensation and working hours, leaves, etc. as per Mines Act-1952
- KU16.** different types of crusher and their importance
- KU17.** different types of crushing principal components, such as gates, chutes and gaps, feeder, conveyor, dust collectors, water sprays etc.
- KU18.** different types of screening equipment like drive mechanism, balance wheel, rocker arms, conveyor belts, guards, etc.
- KU19.** hazards and safety aspects involved in ore processing activities and usage of relevant PPEs
- KU20.** various types of feed arrangement of the crusher

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** note down observations (if any)
- GS2.** read and interpret symbols and readings
- GS3.** read information documents
- GS4.** discuss task lists, schedules and activities
- GS5.** effectively communicate, listen and comprehend the information given by various sources about the site
- GS6.** make decisions pertaining to the concerned area of work
- GS7.** plan and organize the work order and tasks
- GS8.** organize all operation and service manuals so that sorting/ accessing information is easy
- GS9.** detect problems in day to day tasks



Qualification Pack

- GS10.** discuss possible solution with the supervisor for problem solving
- GS11.** make decisions in emergency conditions
- GS12.** follow instructions and work on areas of improvement identified
- GS13.** complete the assigned tasks timely
- GS14.** use reasoning skills to identify and resolve basic problems
- GS15.** analyze and detect any potential problems which could arise during operation
- GS16.** respect persons with disabilities
- GS17.** be sensitive towards all genders

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Perform repair and maintenance of various types of crushers and it's equipments</i>	30	50	-	20
PC1. discuss various types of crusher and their crushing principle	2	3	-	1
PC2. identify the important components of a jaw crusher, cone crusher, etc	2	3	-	1
PC3. check different types of crushing principal components such as gates, chutes and gaps, feeder, conveyor, dust collectors, water sprays etc.	2	3	-	1
PC4. conduct maintenance of the crushing plant	2	2	-	1
PC5. perform daily checking of plant as per check-list followed by organization	2	3	-	1
PC6. change stoke and re- assembly properly	2	3	-	1
PC7. check and replace thrust plate of the crusher	1	3	-	1
PC8. perform replacement of various worn out parts of the crusher	2	3	-	1
PC9. replace main shaft protection sleeve of the crusher	1	3	-	1
PC10. install upper frame assembly of various crusher	2	3	-	1
PC11. replace various worn out parts of vibrating screen, feeder, conveyor etc.	1	3	-	1
PC12. test samples, materials or products to ensure compliance with Indian standard (IS) specifications	2	2	-	1
PC13. conduct operational checks on areas of potential issues like unusual noises/smells, blockages and obstruction, leaks etc. and take corrective actions if required	1	3	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. check indicators that signal need for replacement	2	2	-	1
PC15. replenish coolants, lubricants, fluids and screeners as and when required	1	3	-	1
PC16. follow the safe code of practice for erection, installation, operation, repairs, maintenance, dismantling of plant and ancillary equipments	1	2	-	1
PC17. identify missing or defective components or controls and replace them with genuine OEM recommended components	2	2	-	1
PC18. check oil, fuel tanks for leaks and take necessary actions as per the operational manual	1	2	-	1
PC19. lubricate all the moving parts at regular intervals	1	2	-	2
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N3210
NOS Name	Perform repair and maintenance of crusher and it's equipments
Sector	Mining
Sub-Sector	Engineering Services
Occupation	Mechanical Services
NSQF Level	3
Credits	2
Version	2.0
Last Reviewed Date	08/05/2025
Next Review Date	06/05/2028
NSQC Clearance Date	08/05/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Qualification Pack

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MIN/N3211.Perform installation and preventive maintenance of commonly used mine machineries/equipments	30	50	0	20	100	40
MIN/N3212.Perform troubleshooting and repair of commonly used mine machineries/equipment	30	50	0	20	100	30
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	10
Total	80	130	-	40	250	80

Elective: 1 Underground Metal

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MIN/N1702.Follow Health, Safety and Environmental guidelines for Underground Metalliferous Mines	30	50	-	20	100	20
Total	30	50	-	20	100	20

Elective: 2 Opencast

Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MIN/N1703.Follow Health, Safety, and Environmental Guidelines for opencast mines	30	50	-	20	100	20
Total	30	50	-	20	100	20

Elective: 3 Underground Coal

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MIN/N1704.Follow Health, Safety, and Environmental guidelines for underground coal mines	30	50	-	20	100	20
Total	30	50	-	20	100	20

Optional: 1 Repair-Underground

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MIN/N3209.Troubleshooting and repair/ maintenance of UG mine equipment/machineries	30	50	0	20	100	20
Total	30	50	-	20	100	20

Optional: 2 Crusher

Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MIN/N3210.Perform repair and maintenance of crusher and it's equipments	30	50	0	20	100	20
Total	30	50	-	20	100	20

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
NOS	National Occupational Standard(s)
QP	Qualification Pack
NSQF	National Skills Qualifications Framework
TVET	Technical and Vocational Education and Training
SCMS	Skill Council for Mining Sector
ROPS	Roll-Over Protection Structure
FOPS	Falling Object Protective Structure
PC	Performance Criteria
HEMM	Heavy Earth Moving Machinery
OEM	Original Equipment Manufacturer
DGMS	Directorate General of Mining Safety
MMR	Metalliferous Mines Regulations
CMR	Coal Mines Regulations
SOP	Standard Operating Practice
MVTR	Mine Vocational Training Rules
MRR	Mines Rescue Rules
HSE	health, safety & environment
OB	Over Burden
ERP	Enterprise resource planning
MCDR	Mineral Conservation and Development Rules

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
Sector	A sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	a sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in the industry.
Occupational Standards	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standard	NOS are occupational standards which apply uniquely in the Indian context.

Qualification Pack

Qualification Pack	QP comprises the set of OS, together with the educational, training, and other criteria required to perform a job role. A QP is assigned a unique qualifications pack role
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.